

Pareton: Inference Optimization

Problem

- AI models are being commoditized. As model quality converges, cost and latency become more important competitive advantages.
- Total inference spending is rising because inference demand is growing faster than inference efficiency is improving.
- Inference optimization is a massive search problem. No single team can stay at the frontier alone.

What is Pareton

Pareton is building the optimization layer for AI inference. We continuously discover, validate, and deploy the most efficient way to serve AI workloads across different models, hardware, and deployment environments.

How it works

Engineers submit auditable code contributions (patches to vLLM), targeting specific performance bottlenecks (e.g. KV cache efficiency, prefill scheduling, batching, etc.)

- Submissions are benchmarked automatically across GPU types, context lengths, and request patterns.
- Only improvements that hold across environments earn rewards. Optimizations that work on one GPU config but not another are filtered out.
- The result is a continuously updated, cross-validated optimization baseline organized *per* model.

Why Pareton will win

Inference optimization is a big search problem.

1. There are many components within a serving stack that can be optimized, with even more possible combinations between them.
 - a. *Examples: kernels, KV cache management, batching, scheduling, quantization, speculative decoding, routing, memory management.*
2. The search space also evolves quickly and continuously as new models, hardware, and techniques are introduced, making it difficult for any single provider to stay at the frontier.
3. Most optimization work remains fragmented across individual companies.

Pareton turns optimization into a shared search process where discoveries compound over time. Each new contribution expands the optimization baseline available to future contributors.

We are using **Bittensor** as the incentives layer, aligning hundreds of independent contributors around this single objective.

Business Model

The scalable optimization pipeline with incentivization is the moat.

Near-term: Optimization Pipeline for AI Product Companies (e.g. Cursor)

Chutes: "We spend \$x/year on inference, and are not sure if our stack is optimal since everything is changing so rapidly, and we don't have the energy to continuously benchmark."

Given a customer's inputs (model, workload, hardware, etc.), Pareton continuously discovers and deploys the most efficient way to serve their particular profile, without them requiring a dedicated inference optimization team.

Business Model: Subscription or savings-share agreements. (See: Per-Partner Unit Econ & Revenue Sensitivity → [Google Sheet link](#))

Long-term: Optimization Layer Above Providers

"Which provider, model, config, and serving stack should this workload run on?"

Users do not care whether a request runs on Fireworks, Together, or somewhere else. They care about cost, latency, and quality. Pareton becomes the intelligence layer for inference infrastructure. Given a workload, Pareton determines the optimal provider, model, hardware, and serving configuration. The optimization network continuously generates the knowledge that powers this routing layer.

Business Model: Usage-based routing fees and optimization margins.

Team

Xavier Lyu — Founder [Twitter](#) · [GitHub](#) · [LinkedIn](#)

Xavier started at Pantera Capital as a quant developer, building data pipelines and ML infrastructure, then moved into a fundamental investment analyst role covering protocol economics, market sizing, and competitive dynamics. At the Opentensor Foundation, he worked under Jacob Steeves reviewing subnets, analyzing incentive mechanisms, and identifying potential exploits. He co-designed dTAO tokenomics by running simulations.

Xavier graduated from Cornell University with a Bachelor in Computer Science with honors. He placed in the Platinum Division of the USA Computing Olympiad.

The Ask

Pareton is raising **\$1.4 - 1.6M** to accelerate growth.

Current status: Live on the Bittensor mainnet for the past two months with active miner competition and production infrastructure. The network is currently paused while the protocol is being rebuilt.

Use of funds: Expand the engineering team, build GTM capabilities, and scale adoption among inference providers.

Structure: SAFE with an optional token warrant.

Ideal partners: Strategic investors with experience in AI infrastructure or open-source ecosystems.